

Design Evaluation 2b—XR Podcast Explorer

1. Objective and Validation Metrics

The objective of this evaluation was to determine whether the interaction changes developed after the first evaluation improved first-time users' ability to complete the core XR Podcast Explorer flow with less hesitation and assistance. Five participant records were analysed across the journey from Topic Door -> Episode browsing and selection -> Chapter selection -> Podcast Experience -> Favourite Heart interaction.

Success was evaluated through task completion, time where recorded, errors, hesitation, moderator assistance, direct observation and post-test feedback. Because Participant 1's recording began at Episode browsing, Topic Entry evidence is available for four participants rather than five. A task was treated as independently successful when it was completed without moderator assistance; hesitation and errors were retained as usability evidence even when the task was ultimately completed.

Interaction tested	Validation criterion
Topic Door	Recognise the AI Topic Door as an entry point and enter independently.
Episode browsing	Browse the available Episodes, understand the current/side-card relationship, and select an Episode without prompting.
Chapter selection	Select Interview, recognise the updated Current Chapter Door, and enter the Podcast Experience.
Favourite Heart	Understand, grab and place the Heart in the Favourite Area without assistance, with a clear sense that saving is complete.
Podcast Experience	Recognise that podcast audio is playing and understand the current listening state.
Controller interaction	Use movement, turning, ray selection and grabbing without controller-role confusion.

2. Results

The second round showed a clear improvement in the core spatial flow. Topic entry and Chapter-to-door navigation were consistently successful in the recorded sessions, while Episode browsing was generally completed independently but still produced interaction-expectation mismatches. The Favourite Heart was more understandable than in Evaluation 1, but controller-role conflict and missing completion feedback remained the most important barriers to a confident final interaction.

Area	Result	Key finding
Topic Door	Validated	All four participants with recorded Topic Entry data entered independently (4/4).
Episode Browsing	Validated with refinement needed	All five completed Episode browsing; four completed it without moderator assistance. P1 needed a Trigger prompt, while P2 expected a swipe gesture and P3/P5 showed smaller centre-card hesitation or early selection.
Chapter Selection	Validated	All five reached the Interview chapter and entered the Podcast Experience. The green Current Chapter Door generally communicated the next step; P1 found it useful once noticed.

Area	Result	Key finding
Favourite Heart	Partially Validated	All five completed the interaction, but only three did so independently. P1 and P3 needed assistance because of controller/wayfinding difficulty. P1 and P5 also wanted clearer save confirmation.
Podcast Experience	Partially Validated	Audio was present and noticed in P1; P4 specifically looked for a Now Playing or playback-status cue because the listening state was not visually explicit.
Controller Mapping	Needs Iteration	P1 and P3 reported difficulty combining turning/movement with Heart grabbing and wanted clearer left/right controller responsibilities.

Across the recorded tasks, the strongest evidence is the Chapter interaction: participants understood the relationship between selecting Interview and using the Current Chapter Door. The most repeated remaining problems cluster around the final Favourite interaction and around how Episode browsing communicates its available gestures.

3. Analysis / Insights

Insight 1 - The Chapter-to-door transition has improved substantially

Evaluation 1 identified the Chapter -> Current Chapter Door transition as a major point of uncertainty. In this round, all five participants completed the Chapter task, and P2-P5 did so without assistance. P3 described Chapter selection as the easiest interaction, while P1 reported that the green door helped communicate that it could be opened once the state change was noticed. This indicates that the selected-door state is now functioning as an effective spatial cue, although its visual salience can still be strengthened.

Insight 2 - Episode browsing is usable, but the carousel creates competing interaction expectations

All participants completed Episode browsing, so the interaction is learnable. However, the visual carousel does not always communicate the implemented control. P1 interpreted the centre card as the obvious clickable option because it had stronger pop-up feedback than the side cards. P2 physically moved the controller sideways and expected a swipe gesture. P3 selected the centre card too early, and P5 briefly hovered it before completing browsing. The issue is therefore less about task failure and more about consistency between visual form, hover feedback and expected gesture.

Insight 3 - The Heart metaphor is clearer, but the interaction still has two separate usability problems

Participants generally understood the Heart/Favourite relationship, showing improvement over Evaluation 1. The remaining issues are physical control and completion feedback. P1 and P3 experienced controller-role conflict while holding the Heart, while P1 and P5 were uncertain whether the podcast had actually been saved after placement. These should be treated as separate design problems: one concerns how the user manipulates the object, and the other concerns how the system communicates success.

Insight 4 - Controller responsibilities need to be predictable during object manipulation

P1 and P3 both reported that too many actions felt concentrated on the same controller/hand when the Heart had to be grabbed while the user also needed to move or turn. This increased physical interaction load even

though the intended task was understood. A consistent mapping - for example, left controller for locomotion and right controller for turning, ray selection and grab - would reduce role switching and should be introduced consistently before or within the experience.

Insight 5 - Podcast playback needs a visible listening state

The prototype now includes audio, but audio alone does not fully communicate system state. P4 heard the podcast yet looked for a player or status indicator and suggested a simple Now Playing display. This is not evidence that the audio itself failed; rather, the environment lacks a persistent visual cue showing what Episode/Chapter is currently playing and whether playback has started.

4. Evaluation of Aims

Testing Aim	Evaluation	Evidence
Understand Topic Door	Validated	4/4 participants with recorded Topic Entry data entered independently; P1 Topic Entry was not captured.
Browse/select Episodes independently	Validated	5/5 completed; 4/5 completed without moderator assistance. Remaining issues concern affordance and gesture expectation rather than overall task completion.
Understand Chapter navigation	Validated	5/5 completed the Chapter task; the green Current Chapter Door was generally understood as the next step.
Independently complete Favourite	Partially Validated	5/5 completed, but only 3/5 did so without assistance. P1 and P3 required help during Heart manipulation.
Understand Favourite completion	Partially Validated	P1 and P5 expected clearer confirmation after placing the Heart.
Understand podcast listening state	Partially Validated	Audio was available, but P4 specifically searched for visual playback/status information.
Use controller roles confidently	Partially Validated	P1 and P3 reported controller-role confusion during the Heart task.

Overall, Evaluation 2 validates the core Topic -> Episode -> Chapter -> Podcast Experience structure more strongly than Evaluation 1. The Chapter transition that previously caused uncertainty is now largely resolved. Prototype 2B should therefore focus less on restructuring the content hierarchy and more on interaction polish: Episode carousel behaviour, controller mapping, Favourite completion feedback and playback-state communication.

5. Prototype 2B Concept Iteration

The following Prototype 2B changes are derived directly from the observed problems in this evaluation. Priority is based on repeated evidence and on whether the issue affected independent task completion or confidence in task completion.

Insight / Problem	Prototype 2B Design Response	Priority
Heart manipulation conflicts with movement/turning	Use a consistent two-controller mapping. Keep locomotion on the	High

Insight / Problem	Prototype 2B Design Response	Priority
	left controller and turning/ray/grab responsibilities predictable on the right; provide a brief control cue before object manipulation.	
Favourite save state is unclear	After the Heart reaches the Favourite Area, keep it visibly attached and add immediate confirmation such as a short Saved to Favourites message, glow/animation and/or subtle sound.	High
Favourite Area requires unnecessary movement while holding Heart	Move the Favourite Area closer to the Heart Shelf so the task can be completed with less repositioning.	High
Episode carousel suggests swipe/drag or over-emphasises centre card	Add horizontal swipe/drag browsing if feasible, while retaining ray + trigger as an alternative. Give Prev/Next cards consistent hover/movement feedback so all browsable cards visibly respond.	High
Current Chapter Door state can still be missed	Retain the green selected state and strengthen salience with clearer title/time treatment or a brief Enter selected chapter cue.	Medium
Podcast audio state is not visually explicit	Add a compact Now Playing display showing Episode, Chapter and a simple playback/progress state.	Medium
Successful spatial flow should be preserved	Keep the Topic Door and Current Chapter Door interaction model unchanged except for visual polish, because the door-based hierarchy performed well.	Retain

Prototype 2B should not replace the successful spatial door structure. The iteration should preserve the interaction patterns that participants already understood and target only the points where users hesitated, needed assistance or lacked system feedback.

6. Reflection and Next Steps

Compared with Evaluation 1, this round provides evidence that several earlier design changes were effective. The Chapter-to-door relationship is the clearest example: a transition that previously required explanation was completed successfully across the second-round participant records. This demonstrates the value of using visible spatial state changes rather than adding more conventional menu UI.

The evaluation also shows why task completion should not be treated as the only measure of success. Every participant eventually completed Episode browsing and the Favourite task, but their behaviour revealed different problems: gesture expectation, controller-role conflict and uncertainty about whether a save action had completed. These issues would be hidden by a simple completion-rate measure.

A limitation of the dataset is that P1 Topic Entry was not captured in the available recordings, and P1 did not have recorded task times. Therefore, aggregate claims for Topic Entry use a 4-participant denominator, and timing comparisons should not be made across all five participants. The later cable/app reconnection discussion in P1 was treated as a technical interruption rather than a usability error.

The next Prototype 2B evaluation should test whether the revised Episode interaction is immediately discoverable, whether users can manipulate the Heart without controller-role assistance, whether the Favourite action produces an unmistakable completion state, and whether the Now Playing cue makes the podcast listening state clear. The goal is to retain the validated spatial hierarchy while reducing the remaining interaction-level hesitation.